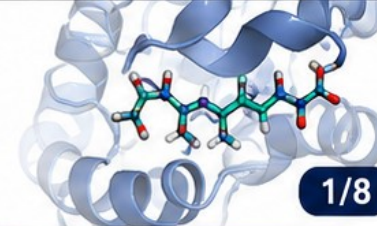


MultiTargetDocking V17.0

What this Colab notebook does and why it matters



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Core purpose



Multi-target, multi-ligand docking workflow



Rigid receptor, flexible ligand



Supports co-crystal validation before production docking



Built for large screens with many proteins and ligands



Resumable runs with checkpoints and audit logs

1



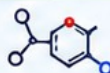
Access gate & Drive setup

2



Receptor input and cleaning

3



Ligand input and preparation

4



Pocket definition / active site selection

5



Optional co-crystal redocking validation

6



Production docking with Vina, Smina fallback

7



Pose clustering, scoring, and QC

8



Organized output packages, reports, and PLIP-ready complexes



Why it is important



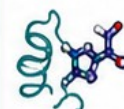
Standardizes docking preparation and execution



Improves confidence using redocking and clustering diagnostics



Produces interpretable CSV, heatmap, HTML, and summary outputs



Creates PLIP-ready protein-ligand complexes for interaction analysis



Organizes results so users can quickly find the best pose per protein-ligand pair



Key scientific note

This workflow performs **rigid-receptor / flexible-ligand docking**. It **does not** run a separate post-generation force-field minimization step.

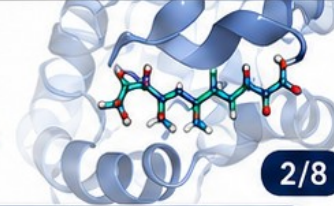


Guide to the notebook settings, workflow, outputs, and result interpretation



High-level controls and run setup

The first settings decide where the run lives, what is executed, and how safely it resumes



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1 Path & run control		
Input	Meaning	Affects
 BASE_DRIVE_DIR	Root folder on Google Drive for the run	Determines where outputs are saved
 RESUME_RUN_DIR	Existing run folder to continue	Reuses a previous run instead of creating a new timestamped run
 FAST_IO_MODE	Use local scratch then sync to Drive	Improves speed for large jobs

2 Workflow selector		
Input	Meaning	Affects
 WORKFLOW_MODE	redock_and_dock / redock_only / dock_only_no_redock	Chooses whether validation, production docking, or both are executed
 REQUIRE_PARSEABLE_DOCKING_SCORE	Require a readable affinity score	Unscored jobs are treated as failed and not ranked

3 Receptor & co-crystal entry		
Input	Meaning	Affects
 RECEPTOR_MODE	mixed_entries or upload_pdbs_to_drive	Controls where receptors come from
 RECEPTOR_ENTRIES_TEXT	List of PDB IDs or receptor entries	Sets which proteins are processed
 COCRYSTAL_SELECTION_MODE	auto / manual_per_protein / mixed	Controls how the reference ligand is chosen
 COCRYSTAL_MANUAL_MAP	Manual mapping of protein to co-crystal ligand	Overrides automatic choice when needed
 COCRYSTAL_FORCE_USE_IF_PRESENT	Keep detected co-crystal even if small	Changes whether small reference ligands are accepted for validation

4 Resumability & safety		
Input	Meaning	Affects
 ENABLE_SQLITE_CHECKPOINTS	Save finished job state in SQLite	Allows resumable Colab sessions
 RESUME_COMPLETED_DOCKING_JOBS	Skip already completed jobs	Saves time on reruns
 VERIFY_CHECKPOINT_INPUT_SIGNATURES	Check that inputs match saved checkpoints	Prevents unsafe reuse after changes
 VERIFY_CHECKPOINT_OUTPUT_FILES	Confirm expected output files still exist	Avoids false completed states
 SAVE_RUN_LOG	Write a persistent run log	Improves traceability and debugging



These settings control **run location, scope, and reliability** before any scientific calculation starts.



Tip: for large campaigns, define the path and resume strategy first.



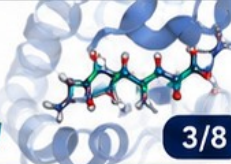


Google Colab

Receptor preparation and pocket definition

These settings change receptor composition, chain selection, and the search space used for docking

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1 Receptor cleaning

Input	Meaning	Affects
KEEP_METALS	Keep metal ions in the receptor	Preserves metal-containing sites, but scoring may be less reliable for metal coordination
KEEP_STRUCTURED_WATERS	Retain selected crystallographic waters	Changes receptor composition and can alter docking contacts
WATER_KEEP_RADIUS	Radius used to keep waters near the co-crystal ligand	Controls how many waters are preserved
PH	Global pH reference	Affects reported protonation/pH context
KEEP_CHAINS	Keep only selected chains	Reduces receptor complexity
REMOVE_NONSTANDARD	Remove nonstandard residues	Simplifies the receptor model
KEEP_HETATM_RESNAMES	Preserve chosen hetero residues such as cofactors	Keeps rigid cofactors in the receptor

2 Chain editing

Input	Meaning	Affects
CHAIN_EDIT_MODE	off or json_map	Enables per-protein chain filtering
CHAIN_REMOVALS_JSON	Mapping such as 2wgj=A	Defines which chains are retained or removed

3 Pocket definition

Input	Meaning	Affects
ENSEMBLE_MODE_WHEN_COCRYSTAL	cocrystal_only / predicted_only / both	Chooses which pockets are docked when a co-crystal exists
N_PRED_WHEN_COCRYSTAL	Number of predicted pockets when a co-crystal is present	Changes how many extra boxes are docked
N_PRED_WHEN_NOCOCRYSTAL	Number of predicted pockets without a co-crystal	Changes screening breadth
FP_CHOOSE_BY	score / drug_score / simple	Controls which predicted pockets are prioritized
ACTIVE_SITE_CONTROL_MODE	auto or json_map	Allows manual override of the docking box
ACTIVE_SITE_OVERRIDES_JSON	Manual center/size JSON	Directly sets the search box for chosen proteins
MANUAL_BOX_SIZE_DEFAULT	Default size for manual boxes	Sets fallback box dimensions

4 Box sizing

Input	Meaning	Affects
FIXED_BOX_SIZE	Fixed cubic box size	Used when automatic scaling is off
AUTO_BOX_SCALING	Scale box from ligand or site geometry	Adapts box size automatically
BOX_MARGIN	Extra padding around the site	Expands the search region
BOX_MIN and BOX_MAX	Lower and upper size limits	Prevents too-small or too-large boxes
INTERACTION_CUTOFF_A	Distance used to identify nearby residues	Affects site inspection and reporting
DOCK_BOX_ANISOTROPIC	Allow per-axis box growth	Makes docking boxes fit elongated ligands better



Pocket definition strongly affects docking quality because it decides **where the ligand is allowed to search**.

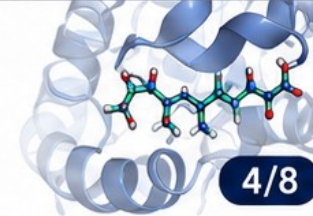


Receptor preparation + pocket setup = the **physical context** for every docking job.



Ligand input and preparation

These settings determine which ligands enter the workflow and how each ligand is prepared for docking



1 Ligand source modes

	LIGANDS_MODE	smiles_inline / upload_sdf_multi / drive_sdf / drive_folder / upload_mol2_pdb / drive_pdbqt	Defines where ligands come from
	INLINE_NAMES_CSV	Names for inline ligands	Labels ligand records
	INLINE_SMILES_CSV	SMILES strings for inline ligands	Creates ligands from text chemistry
	DRIVE_SDF_PATH	Path to a single SDF file on Drive	Imports SDF ligands
	DRIVE_FOLDER_PATH	Path to a folder on Drive	Imports many ligand files
	DRIVE_PDBQT_PATH	Path to ligand PDBQT files	Uses preprepared ligands



2 Filtering and scale control

	LIGAND_MAX	Maximum number of ligands accepted	Caps campaign size
	LIGAND_MIN_HAC	Minimum heavy-atom count	Filters tiny fragments and solvent-like inputs
	LIGAND_MAX_HAC_FOR_DOCKING	Maximum heavy-atom count	Filters overly large ligands
	LIGAND_MAX_MW_FOR_DOCKING	Maximum molecular weight	Limits very large compounds
	SKIP_METAL_CONTAINING_LIGANDS	Skip ligands containing metals	Avoids problematic chemistry
	DOCK_FIRST_FROM_MULTI_SDF	Use only the first molecule from a multi-molecule SDF	Reduces the number of ligands generated



3 Conformers and pH

	CONFGEN_ENABLE	Generate conformers	Increases conformational sampling
	CONFGEN_NCONF	Number of conformers per ligand	Multiplies docking jobs and diversity
	CONFGEN_RMSD	RMSD threshold between conformers	Controls how different conformers must be
	PH_LIST_TEXT	One or more pH values	Generates protonation states at listed pH values
	RESPECT_INPUT_PROTONATION	Keep protonation from uploaded SDF when possible	Preserves curated input chemistry



4 Geometry policy

	PRESERVE_INPUT_3D_GEOMETRY	Keep uploaded 3D coordinates when possible	Protects curated structures
	No separate post-generation force-field minimization	The workflow docks uploaded or generated conformers directly	Important for interpretation of geometry quality
	LIGAND_PREPARATION_POLICY	Input or generated conformers without separate minimization	Explains the preparation philosophy



Ligand source, filtering, conformer count, and protonation settings **directly control** how many docking jobs will be created.



Well-prepared ligands improve interpretability and reduce wasted docking time.

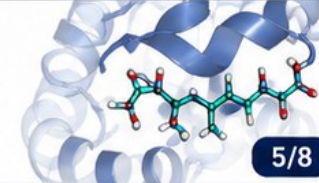




Google Colab

Docking and redocking parameters

These settings trade off speed, sampling depth, and protocol validation strictness



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1 Production docking

EXHAUSTIVENESS	Search effort per job	Higher values search more thoroughly but take longer
NUM_MODES	Number of poses to save	Increases pose diversity and clustering input
ENERGY_RANGE	Score window for saved poses	Controls how many near-best poses are kept
VINA_SEED	Random seed	Affects reproducibility
VINA_CPU	CPU count	Changes runtime and hardware use
REQUIRE_PARSEABLE_DOCKING_SCORE	Require an affinity to keep the job	Prevents rank lists from including incomplete jobs



2 Per-ligand docking box expansion

DOCK_BOX_LIGAND_MARGIN_A	Padding around ligand span	Makes boxes less likely to clip the ligand
DOCK_BOX_MIN_AXIS_A	Minimum axis size	Prevents tiny boxes
DOCK_BOX_MAX_AXIS_A	Maximum axis size	Prevents excessively large boxes
DOCK_BOX_ANISOTROPIC	Grow each axis independently	Better fits long or asymmetric ligands



3 Redocking validation

REDOCK_BOX_POLICY	tight or fixed	Defines the search box used for co-crystal redocking
REDOCK_BOX_MARGIN	Padding used for tight redocking boxes	Expands the validation box
REDOCK_STRATEGY	adaptive_fast / single_fast / thorough	Controls validation rigor and cost
REDOCK_FAST_EXHAUSTIVENESS and REDOCK_FAST_NUM_MODES	First-pass search settings	Fast initial validation attempt
REDOCK_ESCALATION_ENABLED	Allow stronger retries after failure	Improves robustness
REDOCK_ESCALATION_EXHAUSTIVENESS and REDOCK_ESCALATION_NUM_MODES	Retry search settings	Raises search depth after failure
REDOCK_ESCALATION_ATTEMPTS	Maximum extra attempts	Controls retry count
REDOCK_RANDOMIZE_STARTING_POSE	Start redocking from randomized orientations	Makes validation more realistic
REDOCK_REQUIRE_TOP_RANKED_POSE_PASS	Use top-ranked pose as the pass criterion	Makes validation stricter
REDOCK_MIN_PASS_FRACTION	Minimum fraction of attempts that must pass	Decides whether the protocol is accepted
REDOCK_RMSD_FAIL_THRESH	RMSD threshold for failure	Defines the validation cutoff



4 Pocket locking after validation

REDOCK_LOCK_POCKET_MODE	if_pass / always / off	Controls whether the validated redocking pocket is reused for production docking
REDOCK_LOCK_STRATEGY	replace or prepend	Decides how the validated pocket is inserted into the pocket list

Faster**More rigorous**

single_fast



adaptive_fast



thorough



Redocking is important because it **tests whether the docking protocol can recover a known binding mode** before large production screens are interpreted.

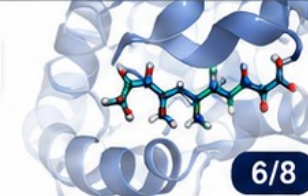


Docking settings govern search depth; redocking settings govern trust.



Analysis, scoring, QC, and reporting

*The notebook does not stop at docking scores;
it also checks pose consistency and prepares user-friendly reports*



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1

Pose analysis and clustering

	CLUSTER_ENABLED	Turn pose clustering on or off	Enables pose-consensus analysis
	CLUSTER_RMSD_THRESH_A	RMSD threshold for grouping poses	Changes cluster granularity
	CLUSTER_MIN_SIZE	Minimum cluster size to keep	Filters very small clusters
	ENABLE_POSE_SCORE_SOFTMIN	Compute a pose-count-normalized soft-min score	Adds a descriptive ensemble score



2

Binding-mode confidence

	CONFIDENCE_THRESH_OCCUPANCY	Low top-cluster occupancy threshold	Flags unstable pose populations
	CONFIDENCE_THRESH_ENTROPY	High cluster entropy threshold	Flags dispersed pose ensembles
	CONFIDENCE_MIN_DG_GAP	Minimum score gap between clusters	Flags weak separation between solutions
	SHOW_CLUSTER_DIAGNOSTICS	Display cluster metrics in outputs	Makes QC easier to review



3

User-facing interpretation thresholds

	AFFINITY_VERY_STRONG_THRESH / AFFINITY_STRONG_THRESH / AFFINITY_MODERATE_THRESH	Plain-language score classes	Affects labels such as very strong, strong, or moderate
	LE_STRONG_THRESH and LE_MODERATE_THRESH	Ligand efficiency interpretation	Helps compare compounds by efficiency per heavy atom
	MW_EFFICIENCY_STRONG_THRESH and MW_EFFICIENCY_MODERATE_THRESH	Docking-score/MW interpretation	Adds size-normalized ranking context
	AFFINITY_STRONG_WARNING_THRESH	Manual-review flag for unusually favorable scores	Encourages caution with extreme scores



4

Reporting and visualization

	ENABLE_HEATMAP	Save a docking-score heatmap	Gives a quick cross-target view
	ENABLE_HTML_REPORT	Build a browsable HTML report	Summarizes best results visually
	ENABLE_PLOTLY_INTERACTIVE_REPORTS	Save interaction plots	Helps exploration of larger datasets
	ENABLE_KEY_FINDINGS_SUMMARY	Write concise narrative summary files	Makes results easier to communicate
	GENERATE_MULTI_TARGET_OVERVIEW_PLOTS	Produce high-level overview plots	Summarizes large screens

Key outputs from this stage:



affinity score



LE



score/
MW



pose_
count



cluster_
count



cluster_
count



top_cluster_
occupancy



cluster_
entropy



cluster_
deltaG_gap



binding_mode_
confidence_class



overall_result_
status



Important: a strong score alone is not enough; the workflow separates **score favorability** from **pose confidence**.



This analysis layer is what turns raw docking into interpretable evidence.



Organized output folders and file map

V17.0 restructures the outputs so users can quickly find protein assets, ligand assets, best poses, and PLIP-ready files



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Why this structure helps



Protein assets are stored once in a canonical protein folder



Ligand assets are stored once in a canonical ligand folder



Each protein-ligand pair has its own folder



00_BEST contains the compact files most users need first



00_START_HERE provides master indexes for large campaigns

RUN/

00_START_HERE/

- OPEN_FIRST.md
- Docking_Result_Index.csv
- PLIP_Ready_Index.csv
- PLIP_INPUT_LIST.txt
- Protein_File_Index.csv
- Ligand_File_Index.csv

01_PROTEINS/

- protein/
 - receptor_raw.pdb
 - receptor_clean.pdb
 - receptor.pdbqt
 - box.txt

02_LIGANDS/

- ligand/
 - confs/
 - prepared_confs.csv

03_DOCKING/

- protein/
 - ligand/
 - 00_BEST/
 - BEST_POSE.pdbqt
 - BEST_POSE.sdf
 - BEST_POSE.pdb
 - PLIP_READY_COMPLEX.pdb
 - BEST_RESULT.json
 - FILES_AND_PATHS.txt
 - 01_JOBS/
 - job folders with raw docking outputs and logs

04_REPORTS/

- Executive_Results_Summary.csv
- best_per_protein_per_ligand.csv
- multidock_rankings_per_job.csv
- Docking_Report.html
- affinity_heatmap.png
- Key_Findings_Summary.md

05_AUDIT/

- dock_job_audit.csv
- receptor_preparation_audit.csv
- ligand_preparation_audit.csv
- cocrystal_redock_audit.csv
- run_manifest.json
- run.log



PLIP-ready outputs



PLIP_READY_COMPLEX.pdb is the protein plus the best docked ligand



PLIP_Ready_Index.csv records where every PLIP-ready file lives



PLIP_INPUT_LIST.txt can be used for batch PLIP runs



This design scales to hundreds of ligands and many proteins with minimal redundancy



The folder organization is designed to be **robust**, **traceable**, and **easy to navigate** in large docking campaigns.

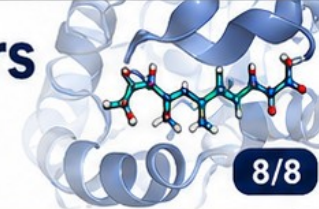


Best-pose files are easy to find; raw jobs remain available without clutter.



How to read the results and why the code matters

The final outputs are intended to support screening decisions, visual inspection, and downstream interaction analysis



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1 Main result files

-  **Executive_Results_Summary.csv** — compact scientific overview of the best results
-  **best_per_protein_per_ligand.csv** — one best-scored row per protein–ligand pair
-  **multidock_rankings_per_job.csv** — complete per-job ranking table
-  **Docking_Report.html** — visual report with pose previews and summaries
-  **Key_Findings_Summary.md** — concise narrative summary
-  **cocrystal_redock_summary.csv** — protocol validation summary

2 Important result columns

	best_affinity_kcal_per_mol	more negative is more favorable, but not an experimental free energy
	ligand_efficiency_le	score normalized by heavy atoms
	docking_score_mw_efficiency	score normalized by molecular weight
	pose_count and cluster_count	how many poses and clusters were observed
	binding_mode_confidence_class	high, moderate, low, or limited sampling
	redock_cocrystal_passed	whether validation passed
	overall_result_status	final interpretation / required caution

3 Recommended user workflow

-  Open 00_START_HERE and read the index CSV files
-  Review Docking_Result_Index.csv or Executive_Results_Summary.csv
-  Go to 03_DOCKING/protein/ligand/00_BEST for the best pose files
-  Use PLIP_READY_COMPLEX.pdb with PLIP to analyze interactions
-  Consult the HTML report, heatmap, and audit files for broader review

4 Why this code is important

-  Automates a complete docking pipeline in one Colab notebook
-  Improves reproducibility with checkpoints and organized outputs
-  Scales from small tests to large multi-protein / multi-ligand screens
-  Separates raw docking scores from confidence and validation evidence
-  Makes downstream structural interpretation faster and easier



Input setup (Drive)



Docking runs (Vina, Smina)



Checkpoints & validation



Organized outputs & reports



Results review & decisions



Best practice: prioritize ligands that combine **favorable scores**, **acceptable LE**, **strong pose confidence**, and **successful co-crystal redocking** when available.



The notebook is valuable because it turns complex docking runs into organized, review-ready evidence.



<https://colab.research.google.com/drive/11RcacPSMRs-TiL-BVTMKI-hBFuIuGPmE?usp=sharing>